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ORIGINAL PAPER

Multivariate multiscale complexity-entropy causality plane analysis for complex time series

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Abstract The multivariate multiscale complexity-entropy causality plane (MMCECP) is introduced for evaluating the dynamical complexity and long-range correlations of multivariate nonlinear systems. Numerical simulations from different classes of systems are applied to confirm the effectiveness of the proposed measure. We observe that the MMCECP not only can characterize the deterministic properties of the systems, but also can distinguish Gaussian and non-Gaussian processes. Moreover, it is immune to varying degrees of noises at large scales. Then we apply it to financial time series analysis, mainly investigating the classification of stock market dynamics. Empirical results illustrate that the MMCECP is robust and valid to detect the physical structures of stock markets.

Keywords Complexity-entropy causality plane · Multivariate · Multiscale · Permutation entropy · Nonlinear time series

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1 Introduction

Complexity of time series generated by nonlinear dynamical systems is a hot topic in scientific fields [1]. In last several years, a variety of complexity measures have been introduced, such as Kolmogorov complexity [2], fractal dimensions [3] and Lyapunov exponents [4]. Moreover, dynamics entropy has been increasingly used to characterize the complexity of nonlinear systems [5,6]. As an information measure that quantifies the uncertainty of systems, researchers have developed many kinds of entropy methods, including approximate entropy and sample entropy [7,8]. Even though these methods have been widely used to different fields and some new definitions of complexity measures are being motivated, there are still some open and subtle issues to be solved. Bandt and Pompe first observed that most of the complexity methods rely on specific algorithms, and sometimes, there may also exist tuning parameters. As a direct effect, the results might be doubtful without knowing the details of the methods or selecting the proper parameters.

With regard to the problems above, Bandt and Pompe proposed permutation entropy (PE) by means of comparing neighboring values that could be suitable for any time series [9]. The ordinal pattern is achieved by first the reconstruction of underlying phase space and then comparison of consecutive or non-consecutive values. In addition, its relative probability density functions (PDF) is unvarying in terms of nonlinear monotonous transformations. The algorithm of this technique is simple and it has the ability to distinguish among periodic, chaotic and stochastic signals, even the analyzed data are in the company of noise. It is extensively utilized in nonlinear dynamic systems [10–12]. In particular, combining permutation entropy with statistical complexity measure proposed by López et al. [13], Rosso et al. firstly introduced complexity-entropy causality plane (CECP) to further detect the underlying complexity of dynamical systems [14–17]. This approach integrates entropy and nonlinear dynamic concepts and can extract the temporal structure of the time series according to the ordinal pattern's PDF. And it has been successfully applied to widespread scientific areas for distinguishing noise, chaotic system and stochastic process [18–24].

Here we extend complexity-entropy causality plane into multivariate systems. Although Ribeiro et al have improved the algorithm for higher-dimensional patterns proved to be promising for recognizing between two-dimensional patterns [24], this new method has some shortages; for example, it can work well in two-dimensional systems but may not be appropriate for three-dimensional or higher-dimensional systems. Furthermore, the higher the dimension of a concerned system is, the more complex the procedure is and the larger the number of variates needed is. In this paper, there are no extra variates in the extension of CECP. What should be done is just calculating the multivariate permutation entropy of time series at first and then just follow some similar steps [25–27]. To deserve to be mentioned, to better describe the intrinsic properties and physical structures of multivariate system, we consider the multiscale coarse-grained method and introduce the multivariate multiscale complexity-entropy causality plane (MMCECP).

The rest of the paper is organized as follows. Section 2 introduces the methodologies of statistical complexity measure and multivariate permutation entropy. Then we integrate these measures and present MMCECP. In Sect. 3, we first select ten different kinds of processes to validate the effectiveness of the proposed method, and then add varying degrees of noises to chaotic maps to detect the influence of noise on results. Section 4 displays the empirical application in financial time series. Finally, the conclusions are drawn in Sect. 5.

2 Methodology

2.1 Statistical complexity measure

Statistical complexity measure is defined to quantify the complexity of a system with a simple structure but complex dynamical characteristics, which can also extract implicit patterns [28]. In addition, the statistical complexity involves that there are two opposing extremes in the nonlinear dynamic system, that is, completely order and maximum randomness. In both situations, its structure is definitely simple, so the statistical complexity is null. However, there exist plenty of possible physical structures between these two extreme situations, which can be mirrored by the underlying system probability distribution properties [29].

Considering a given nonlinear system with any arbitrary discrete probability distribution $P = \{p_i, i = 1, 2, \ldots, N\}$ with N possible states, the classical Shannon information theory is described as [5]:

$$S[P] = -\sum_{i=1}^{N} p_i \log(p_i) \tag{2.1}$$

The values of $S[P]$ quantify the uncertainty and irregularity of the system with some degree. If $S[P] = 0$, we can certainly forecast that all the possible matters i with probability p_i will actually occur. On the contrary, the entropy value reaches maximum when the distribution is uniform, that is, $S[P_e] = S_{\max} = \log N$ where $P_e = \{1/N, 1/N, \ldots, 1/N\}$.

Another complexity measure is defined as "disequilibrium," denoted by Q, which depicts the distance between a specified probability distribution and the uniform probability distribution. The disequilibrium Q can be described with respect to the extensive Jensen–Shannon divergence [16]:

$$\mathcal{Q}_J[P, P_e] = \mathcal{Q}_0 \mathcal{J}[P, P_e] \tag{2.2}$$

where $\mathcal{J}[P, P_e]$ is Jensen–Shannon divergence mentioned above with

$$\mathcal{J}[P, P_e] = S[(P + P_e)/2] - S[P]/2 - S[P_e]/2 \tag{2.3}$$

and $\mathcal{Q}_0$ is a normalization constant, equal to the inverse of maximum possible value of $\mathcal{J}[P, P_e]$. Therefore, the statistical complexity measure is formed through the product

$$C_{JS}[P] = \mathcal{Q}_J[P, P_e]\mathcal{H}_S[p] \tag{2.4}$$

of the normalized Shannon entropy

$$\mathcal{H}_S[p] = S[p]/S_{\max} \tag{2.5}$$

The statistical complexity measure reflects the interrelationship between the quantity of information and its disequilibrium in the dynamic system.

2.2 Multivariate permutation entropy

In order to estimate the two quantifiers mentioned before, a related probability distribution should be manufactured in advance. Bandt and Pompe proposed a successful approach to characterize probability distribution considering the time causality. Based on the theory of phase space reconstruction, the Bandt–Pompe method, named permutation entropy (PE), only takes temporal structure of time series into consideration by means of ordinal structure. Moreover, ordinal pattern distribution is constant in terms of nonlinear monotonous transformation [9]. Here we mainly introduce multivariate permutation entropy which can analyze the cross-channel variability of multivariate time series [25,26].

Given a multivariate time series $\{x_{i,t}\}_{t=1}^{N}$, $i = 1, 2, \ldots, m$, where m represents the number of variates and N denotes the number of samples in each variate, an embedding dimension d and time delay τ, consider the ordinal patterns of order d generated by

$$l_{\text{l}} \to X_{i,l}^{d,\tau} = (x_{i,l-(d-1)\tau}, x_{i,l-(d-2)\tau}, \ldots, x_{i,l-\tau}, x_{i,l}) \tag{2.6}$$

for $l = 1, 2, \ldots, N-(d-1)\tau$ and $i = 1, 2, \ldots, m$. For each of these $N-(d-1)\tau$ vectors, we explore the permutations $\pi = (r_0, r_1, \ldots, r_{d-1})$ of $(0, 1, 2, \ldots, d-1)$, denoted by

$$x_{i,l-r_{d-1}\tau} \le x_{i,r_{d-2}\tau} \le \ldots \le x_{i,l-r_1\tau} \le x_{i,l-r_0\tau} \tag{2.7}$$

Then for all the $d!$ possible permutations π of order d, the associated relative frequencies $p_{i,j}$ can be calculated by

$$p_{i,j} = \frac{\sharp\{l \le L, \text{type}(X_{i,l}^{d,\tau}) = \pi_j^{d,\tau}\}}{L} \tag{2.8}$$

where the symbol $\sharp$ represents the cardinality of the times of permutation $\pi_j^{d,\tau}$, $L = N-(d-1)\tau$ and $j = 1, 2, \ldots, d!$. Then a matrix $P(m, d!) = \{p_{i,j}\}$ is obtained mirroring the distribution of the motifs in multivariate time series. It holds $\sum_{i=1}^{m}\sum_{j=1}^{d!} p_{i,j} = 1$.

In accordance with this algorithm, the original multivariate time series converted into a time-dependent matrix where the related statistics and entropies can be clearly drawn out. Meanwhile, the marginal relative frequencies illustrating the distributions of the motifs can be calculated as $p_{\cdot j} = \sum_{i=1}^{m} p_{i,j}$, for $j = 1, 2, \ldots, d!$. Thus, the multivariate permutation entropy (MPE), demonstrating cross-channel complexity, can be evaluated by

$$H_{\text{MPE}} = -\sum_{j=1}^{d!} p_{\cdot j} \log p_{\cdot j} \tag{2.9}$$

It is obvious that the probability distribution P can be obtained according to the fixed embedding dimension d and the embedding delay time τ. The former parameter is greatly crucial to estimate the suitable probability distribution since it influences the quantity of accessible states, denoted by $d!$. Furthermore, the length of time series N must satisfy the condition $N \gg d!$ to obtain an authentic statistics. In particular, Bandt and Pompe consider the embedding dimension d within the range of [3, 7]. As for the option of the other parameter, they recommend the time delay $\tau = 1$ specifically [30].

2.3 Multivariate multiscale complexity-entropy causality plane

In order to better describe the intrinsic properties of multivariate system, multivariate multiscale complexity-entropy causality plane (MMCECP) is proposed in this paper. Given a m-dimensional time series $\{x_{i,t}\}_{t=1}^{N}$, $i = 1, 2, \ldots, m$, the procedure of MMCECP algorithm consists of three steps, as follows.

Step 1: For a scale factor s, we obtain the coarse-grained m-dimensional time series $Y_k^s = y_{i,k}^{s}{}_{i=1}^{m}$ by

$$y_{i,k}^{s} = \frac{1}{s}\sum_{t=(k-1)s+1}^{ks} x_t^i \tag{2.10}$$

where s denotes the scale factor and $1 \le k \le \lfloor \frac{N}{s} \rfloor$ [31] (Fig. 1).

Step 2: Given the embedding dimension d and time delay τ, calculate permutation entropy $H_{\text{MPE}}(s)$ of coarse-grained multivariate time series $\{Y_k^s\}$ with different scale factors s and then normalize the value $H_{\text{MPE}}(s)$ by

$$H_{\text{NMPE}}(s) = H_{\text{MPE}}(s)/\log(d!) \tag{2.11}$$

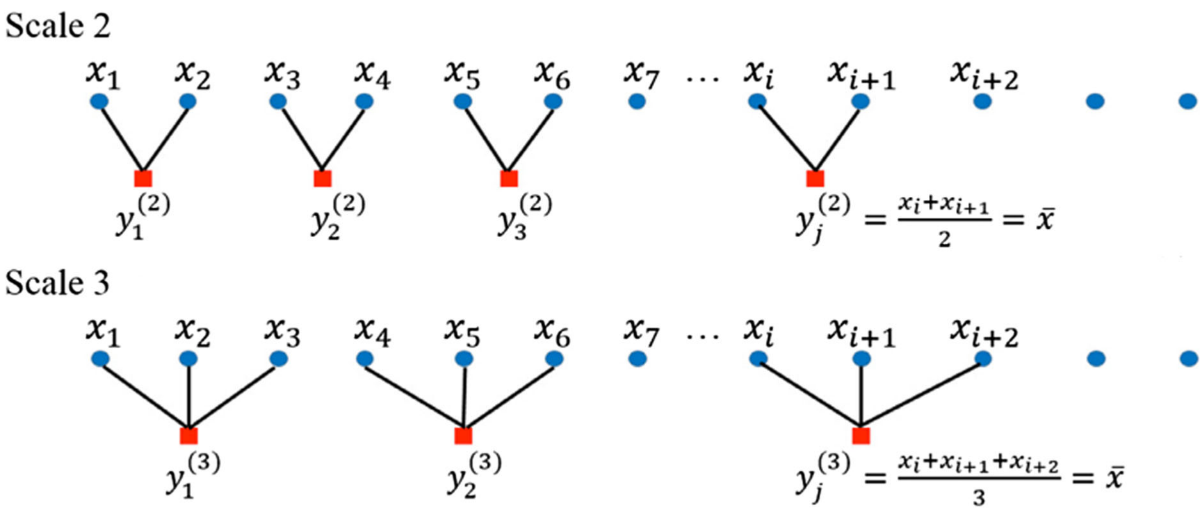

Fig. 1 Schematic diagram of the coarse-graining procedure when the scale are 2 and 3, respectively

Step 3: Compute the new complexity measure $C_{MM}[P]$:

$$C_M[P] = \mathcal{Q}_M[P, P_e]\mathcal{H}^{(s)}_{\text{MPE}}[p] \tag{2.12}$$

with $\mathcal{Q}_M[P, P_e] = \mathcal{Q}_0 \mathcal{J}_M[P, P_e]$ and $\mathcal{Q}_0 = -2\{(\frac{d!+1}{d!}\log(d!+1) - 2\log(2d!) + \log(d!)\}^{-1}$.

To estimate the time evolution of $C_M[P]$, a plot of $C_M[P]$ versus H_{NMPE} can be applied, in the name of MMCECP. In this situation, H_{NMPE} can be considered as an arrow of time [32]. Locating the chaotic and stochastic processes at distinct planar positions, it is more obvious to discriminate them by the representation space especially more complex multivariate systems. Likewise, for a certain H_{NMPE}, there exists a range of possible $C_M[P]$ values between a minimum $C_{\min}$ and a maximum $C_{\max}$ [33].

3 Numerical simulations

In this section, we display several different classes of systems: chaotic systems, stochastic process and periodic process, to illustrate the effectiveness of the proposed MMCECP.

(a) Chaotic systems

(1) Hénon map [34]

$$\begin{aligned} x_{k+1} &= 1 - a x_k^2 + y_k \\ y_{k+1} &= b x_k \end{aligned} \tag{3.1}$$

Parameter value: $a = 1.4$ and $b = 0.3$; initial condition: $x_0 = 0$ and $y_0 = 0.9$.

(2) Burgers map [35]

$$\begin{aligned} x_{k+1} &= a x_k - y_k^2 \\ y_{k+1} &= b y_k + x_k y_k \end{aligned} \tag{3.2}$$

Parameter value: $a = 0.75$ and $b = 1.75$; initial condition: $x_0 = -0.1$ and $y_0 = 0.1$.

(3) Tinkerbell map [22]

$$\begin{aligned} x_{k+1} &= x_k^2 - y_k^2 + a x_k + b y_k \\ y_{k+1} &= 2 x_k y_k + c x_k + d y_k \end{aligned} \tag{3.3}$$

Parameter value: $a = 0.9$, $b = -0.6$, $c = 2.0$ and $d = 0.5$; initial condition: $x_0 = -0.1$ and $y_0 = 0.1$.

(4) Gingerbreadman map [36]

$$\begin{aligned} x_{k+1} &= 1 - y_k + |x_k| \\ y_{k+1} &= x_k \end{aligned} \tag{3.4}$$

Initial condition: $x_0 = 0.5$ and $y_0 = 3.7$.

(5) Lorenz system [37]

$$\begin{cases} \dot{x} = s(y - x) \\ \dot{y} = rx - y - xz \\ \dot{z} = xy - bz \end{cases} \tag{3.5}$$

Parameter value: $s = 16$, $r = 45.92$ and $b = 4.0$; initial condition: $x_0 = -1$, $y_0 = 0$ and $z_0 = 1$.

(6) Rössler system [38]

$$\begin{cases} \dot{x} = -y - z \\ \dot{y} = x + ay \\ \dot{z} = b + z(x - c) \end{cases} \tag{3.6}$$

Parameter value: $a = 0.2, b = 0.2$ and $c = 5.7$; initial condition: $x_0 = -1,\ y_0 = 0$ and $z_0 = 1$.

(7) Chen system [39]

$$\begin{cases} \dot{x} = a(y - x) \\ \dot{y} = (c - a)x - xz + cy \\ \dot{z} = xy - bz \end{cases} \tag{3.7}$$

Parameter value: $a = 35,\ b = 3$ and $c = 28$; initial condition: $x_0 = -1,\ y_0 = 0$ and $z_0 = 1$.

(b) Stochastic process

(8) Noise with f^{-k} [16]
The f^{-k} power noise is generated as follows: (i) produce random number in the interval $(-0.5, 0.5)$ by the RAND function of MATLAB; (ii) the fast Fourier transform (FFT) y_k^1 is obtained and y_k^2 equals to $y_k^1 \cdot pf^{-k/2}$; (iii) y_k^2 is symmetrized in order to obtain a real function, and then, the pertinent inverse FFT x_i is the expected k-noise. Here we generate three independent f^{-k} time series, y_1^k, y_2^k and y_3^k, and then construct three-dimensional time series, $Y = [y_1^k, y_2^k, y_3^k]$.

(9) Three-dimensional Gaussian white noise
Here we generate a trivariate Gaussian white noise where three data channels are mutually independent, namely $G = [g_1, g_2, g_3]$.

(c) Periodic process

(10) Three-dimensional time series generated by function sin$t$
In order to obtain periodic time series, we generate three independent signals $x_i = \sin(0.2\pi t) + b_i, i = 1, 2, 3$ with same period and different amplitudes. Set $b_1 = 0$, $b_2 = 0.5$ and $b_3 = -0.5$. Finally, the three-dimensional periodic time series is $X = [x_1, x_2, x_3]$.

Figure 2 exhibits, for all the different kinds of signals (chaotic, stochastic and periodic) here considered,

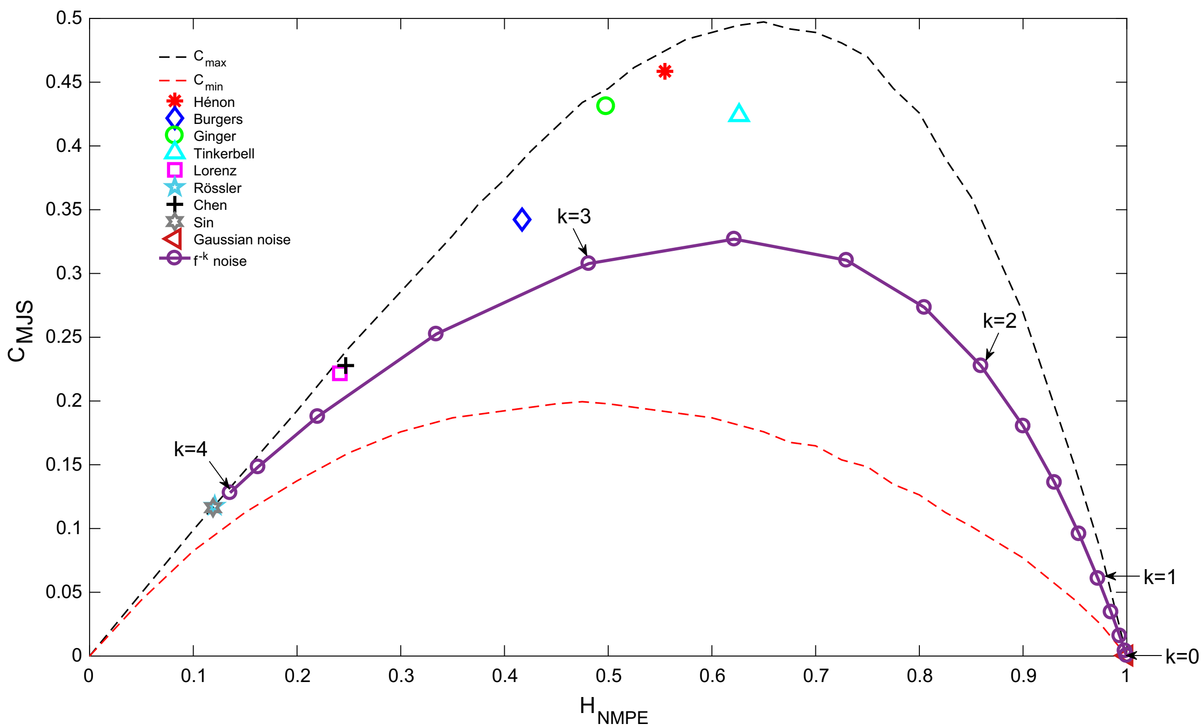

Fig. 2 Localization of different kinds of signals in the multivariate complexity-entropy causality plane. Dash lines stands for curves of minimum $C_{\min}$ and maximum $C_{\max}$, respectively

the multivariate complexity-entropy causality plane location. The parameters of multivariate permutation entropy are estimated for $d = 6$, $\tau = 1$, and the length of each channel of signals is $N = 10^5$ for a plateau of results. In addition, the stochastic process, noise with f^{-k} power spectrum, is shown in the planar localization ($0 \leq k \leq 4$, with $\Delta k = 0.25$), and the corresponding values are averaged for ten simulations. Without loss of generality, almost all the points related to the signals mentioned above are discovered near to the maximum complexity curve $C_{\max}$. And the periodic process has less complexity as expected. It is surprising to find that the location of some of the chaotic maps, Gingerbreadman map and Rössler system, is in the vicinity of periodic process applying the proposed multivariate CECP measure, which is little biased against the results presented in the previous study [16,20]. Thus, the multiscale research for multivariate signals is extremely necessary. For the stochastic processes, there is no doubt that the uncertainty of multivariate Gaussian white noise is the largest and that the complexity is the least and the location is greatly close to the extreme value $(H_{\text{NMPE}}, C_{\text{MJS}}) = (1, 0)$. In terms of the planar representations of f^{-k} power spectrum, the entropy behavior witnesses a downward trend with the increase of k from $H_{\text{NMPE}} \sim 1$ for $k = 0$ to $H_{\text{NMPE}} \sim 0.14$ for $k = 4$, in accordance with the unidimensional case, indicating that correlated colored noises exhibit less randomness for $k > 0$. The related statistical complexity values see an upward trend at first before reaching the highest point (around 0.35) and then gently decrease within the range of $C_{\min}$ and maximum $C_{\max}$.

Figure 3 illustrates the change of C_{MMJS} and H_{NMPE} with different scales s as well as the MMCECP for seven chaotic maps previously displayed. And the maximum coarse-grained scale is $s = 60$. Each row corresponds to a specific chaotic map. In general, the normalized multivariate permutation entropy H_{NMPE} witnesses an increasing tendency to maximum with different degrees for all the chaotic maps, representing that the correlations in the reconstruction of coarse-grained signals vanish. Meanwhile, the values of C_{MMJS} see an opposite trend for all the other six chaotic maps except R*ö*ssler system in varying degrees, implying the loss of dynamical structures as the scale grows up. And the values of points in almost all the multivariate multiscale CECP rise at first and then bottom out close to the extreme point after reaching the peak. This phenomenon indicates the irregular and random motions with lower determinacy in most chaotic maps. In particular, H_{NMPE} values of H*é*non map soar to the corresponding maximum values at small scales and then hold the line, which indicates the randomness in larger scales. And it is worth noticing that the three curves for R*ö*ssler system grow up steadily with the increase in scale. The reason is that the system similar to Lorenz system is relatively stable and easier to analyze quantitatively.

Summing up, as the scale increases, the sample interval length becomes larger, enhancing the loss of correlations in the coarse-grained time series. Therefore, the lower complex dynamics shows up at large scales. And according to comparing the different levels of downward trend in the graphs of each column, we can distinguish various chaotic maps, making multiscale analyses necessary and effective.

In the next part, we add different levels of Gaussian white noise to four typical chaotic maps, Hénon map, Burgers map, Lorenz system and Rössler system, to identify the influence of noise on multivariate multiscale CECP. The standard deviations of noise vary from $\sigma = 0.025$ to $\sigma = 0.15$, with step of $\Delta = 0.025$. The related results are the average value of ten observations.

Figure 4 shows the changes in multivariate multiscale CECP for four typical chaotic maps, Hénon map, Burgers map, Lorenz system and Rössler system, after adding different levels of Gaussian white noises. For the sake of simplicity, the maximum scale is set $s = 20$ and other parameters maintain constant. Generally speaking, the four chaotic maps are sensitive to noise at small scales especially Lorenz system, in which system the values fluctuate more violently as the degree of noise added increases. On the contrary, the fluctuations of R*ö*ssler system are much more slight, confirming the fact that the system is more stable and robust to noise by analyzing three variates of system [40]. Likewise, they are all immune to various levels of noise at larger scales despite some fluctuations.

4 Analyses of financial time series

4.1 Description of financial time series

In order to illustrate the utilization of the proposed multivariate multiscale CECP in practical cases, we apply this method to stock markets to characterize the com-

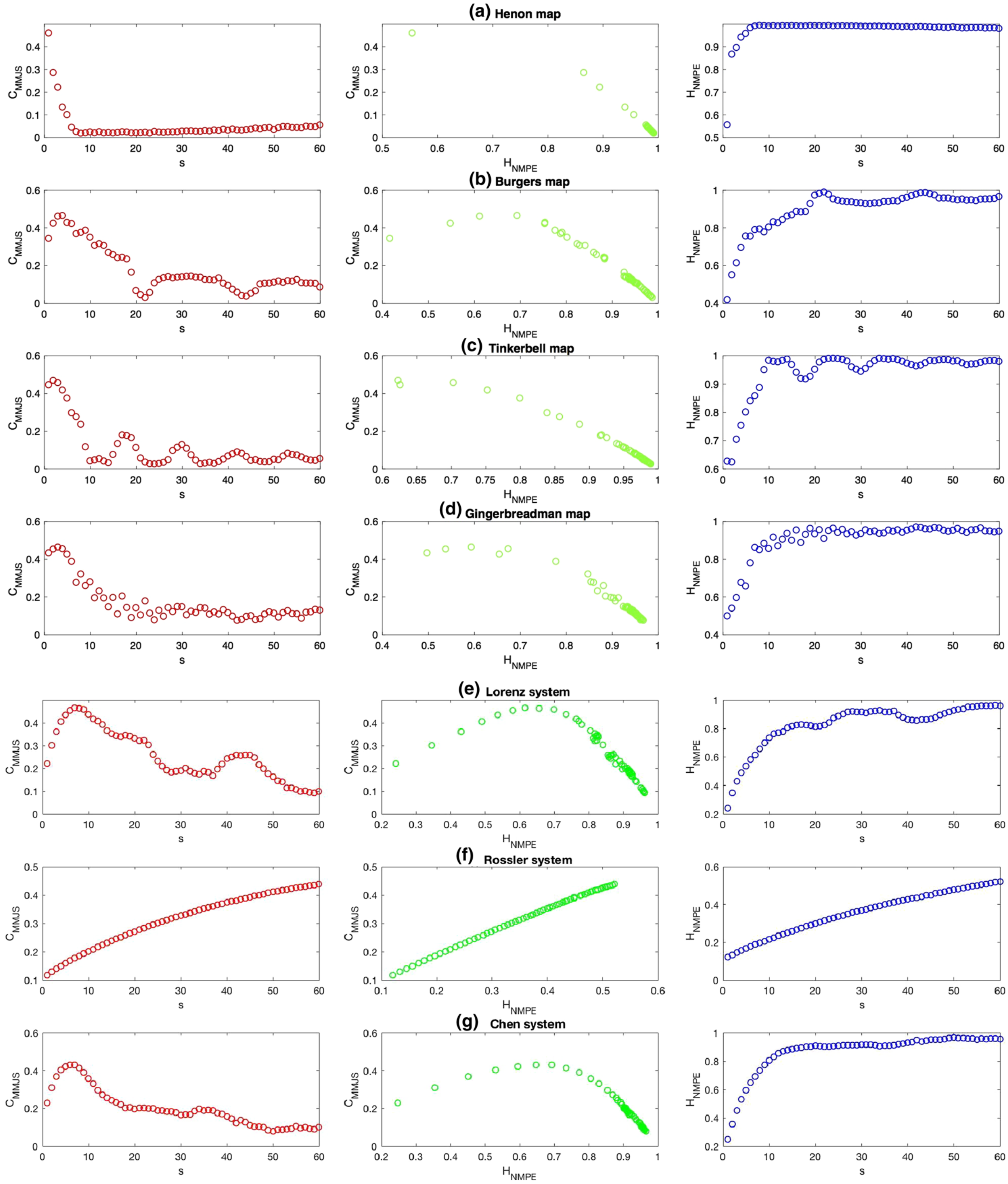

Fig. 3 C_{MMJS} and H_{NMPE} versus scale s and MMCECP for seven chaotic maps

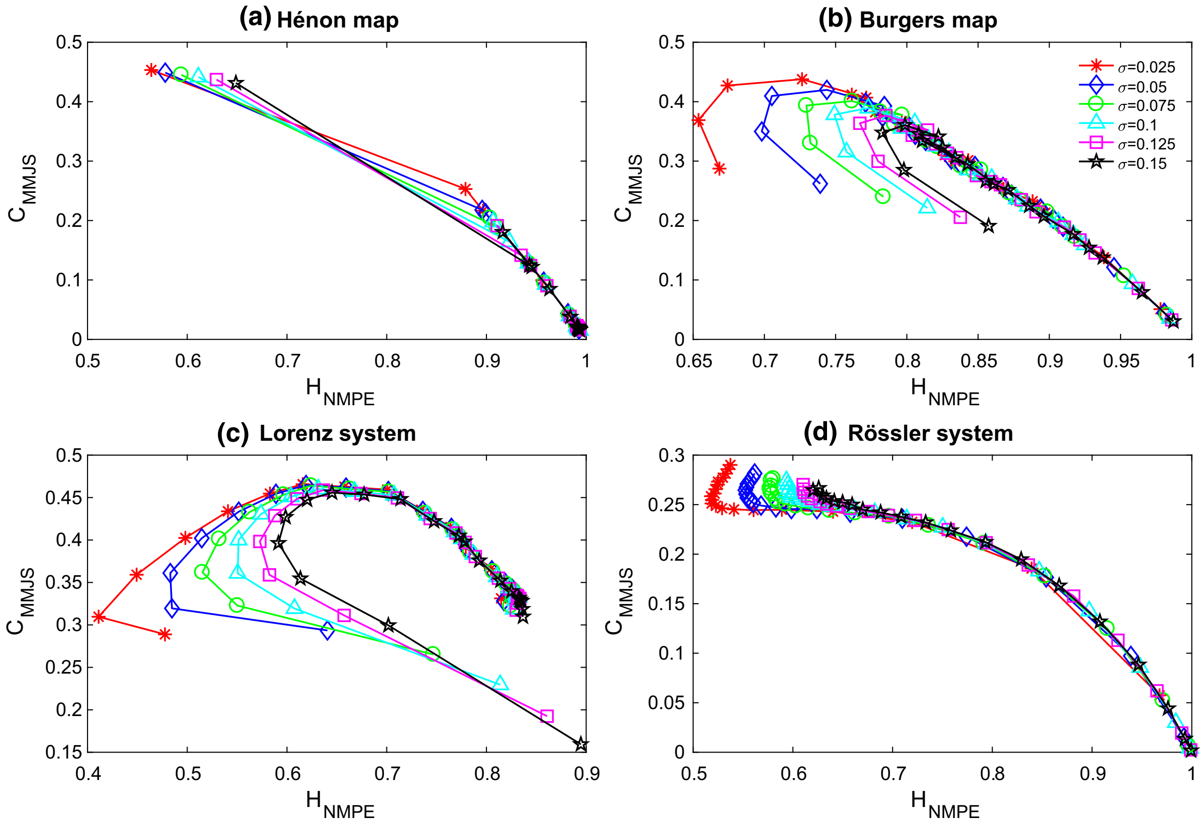

Fig. 4 Localization of four typical chaotic maps in the multivariate multiscale complexity-entropy causality plane with different levels of noises added: **a** Hénon map; **b** Burgers map; **c** Lorenz system; **d** Rössler system

plexity and correlations among them [41]. Here, we employ the daily closing price and trading volume of 21 stock indices from January 1, 2004, to June 30, 2017. These stock indices are divided into three different regions: Americas, Europe and Asia and Pacific. In addition, there are fourteen developed and seven emerging stock indices listed. The classification refers to Morgan Stanley Capital Index (MSCI) methodology (www.msci.com). The datasets are collected from the Web site http://in.finance.yahoo.com/. In the experiment, the detailed information of 21 stock indices is shown in Table 1. And the average of realizations is 3346.

Since financial time series is non-stationarity, we eliminate the unwanted data and then recombine the time series for the sake of the synchronicity [42]. In this section, we adopt logarithmic price difference given by:

$$x_n = \log(S_n) - \log(S_{n-1}) \tag{4.1}$$

where S_n means the closing price of nth trading day.

In terms of trading volume, we first standardize the data to keep the uniformity. Then we also use the logarithmic trading volume difference to deal with the standardized data so as to decline the diversity between them.

To find the optimum time delay of multivariate financial time series, the plots for multivariate permutation entropy H_{NMPE} and statistical complexity measure C_{MJS} as a function of the time delay τ are displayed in Fig. 5. Considering that there are 21 stock indices in total, they are shown by three different regions: Americas, Europe and Asia and Pacific. Results imply that these stock indices can be effectively distinguished when the time delay τ equals to 1. However, as the time delay τ increases, the values of H_{NMPE} and C_{MJS} mix together with fluctuations and it is difficult to identify them. Thus, the optimum time delay of the financial time series is $\tau = 1$.

Table 1 World stock indices

Region	Country	Symbol	Company	Number of observations
Americas	1. Brazil (E)	BVSP	Bovespa Index	3341
	2. Canada (D)	GSPTSE	S&P/TSX Composite Index	3387
	3. Mexico (E)	MXX	IPC Mexico Index	3380
	4. USA (D)	DJI	Dow 30 Index	3398
	5. USA (D)	GSPC	S&P 500 Index	3398
	6. USA (D)	IXIC	Nasdaq Index	3398
	7. USA (D)	NYA	NYSE Composite Index (DJ)	3398
Europe	8. Austria (D)	ATX	ATX Index	3152
	9. Findland (D)	OMXHPI	OMX Helsinki Index	3387
	10. France (D)	FCHI	CAC 40 Index	3351
	11. Germany (D)	GDAXI	DAX Index	3432
	12. Sweden (D)	OMXS 30	OMX Stockholm 30 Index	3385
	13. Switzerland (D)	SSMI	SMI Index	3395
Asia and Pacific	14. Australia (D)	AORD	ASX All Ordinaries Index	3278
	15. China Mainland (E)	SSEC	Shanghai Composite Index	3300
	16. China Hongkong (D)	HSI	Hang Seng Index	3316
	17. China Taiwan (E)	TWII	TSEC Weighted Index	3340
	18. India (E)	BSESN	BSE Sensex Index	3320
	19. Indonesia (E)	JKSE	Jakarta Composite Index	3295
	20. Japan (D)	N225	Nikkei 225 Index	3283
	21. Korea (E)	KS11	KOSPI Composite Index	3358

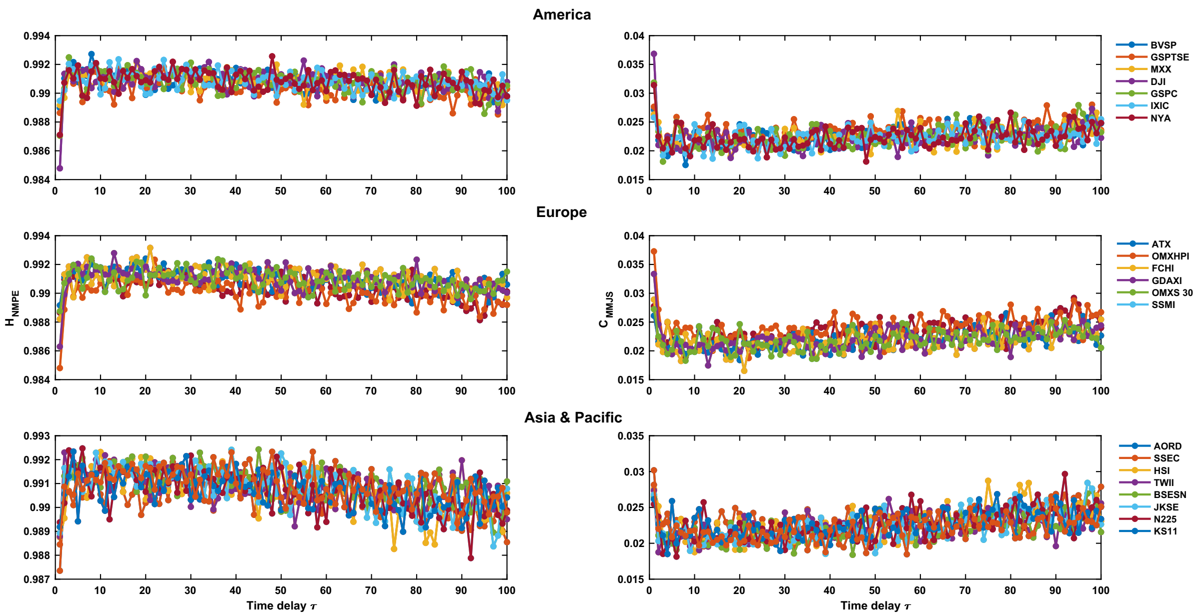

Fig. 5 Change of C_{MJS} and H_{NMPE} as a function of time delay τ

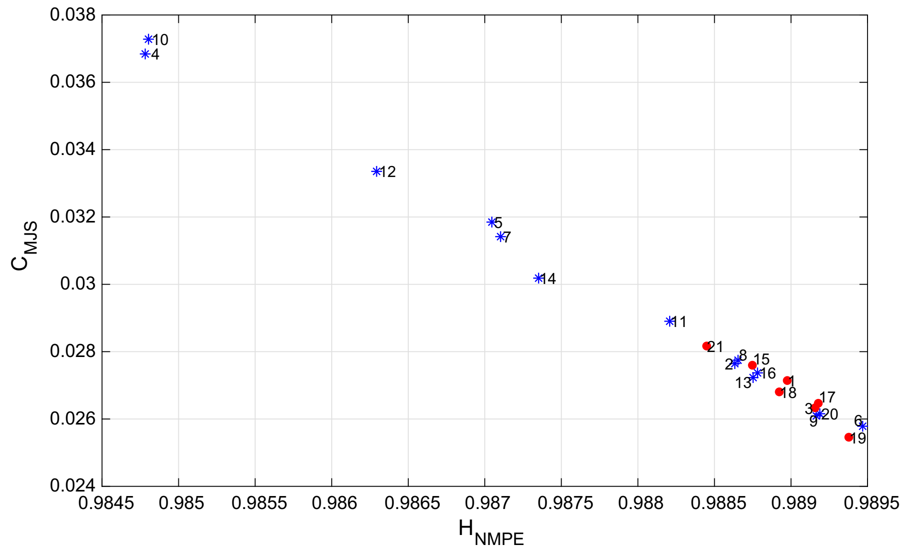

Fig. 6 Localization of 21 stock indices from different regions in the multivariate multiscale complexity-entropy causality plane

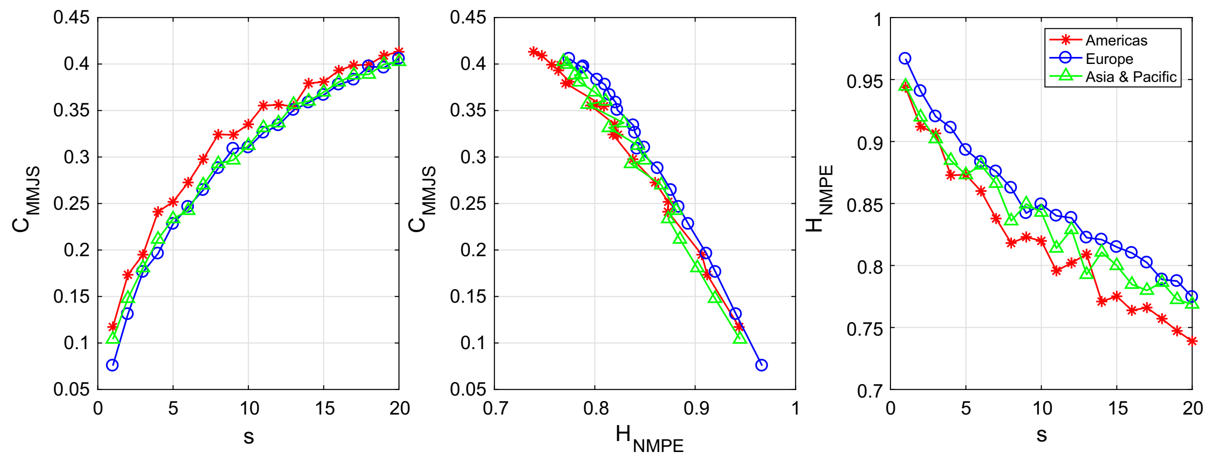

Fig. 7 Multiscale analysis of H_{NMPE} and C_{MMJS} averaged over all the stock indices of each region: Americas, Europe and Asia and Pacific

The localization of different stock indices in the complexity-entropy causality plane is plotted in Fig. 6. The parameters of embedding dimension and time delay are $d = 6$ and $\tau = 1$, respectively ($3346 \gg 6! = 720$). We find that the stock indices of emergent countries have higher entropy and lower complexity than those of developed countries, in the vicinity of the extreme point $(H_{\mathrm{NMPE}}, C_{\mathrm{MJS}}) = (1, 0)$. Conversely, the complexity of stock indices from developed countries is more significant, particularly those from Americans and Europe like USA, Germany and Sweden. Consequently, the values of H_{NMPE} are relatively high, mostly due to the combination of closing price and trading volume of each of the stock indices.

It is of great importance to notice that only six successive days are considered here to construct ordinal patterns. However, the multivariate permutation probability distribution related to each of the stock indices provides the information corresponding to the long-range correlations. In other words, different probability distribution determines different degrees of long-range correlations. Therefore, it is essential to analyze the multiscale characters of 21 stock indices given in Table 1. The maximum scale is $s = 20$.

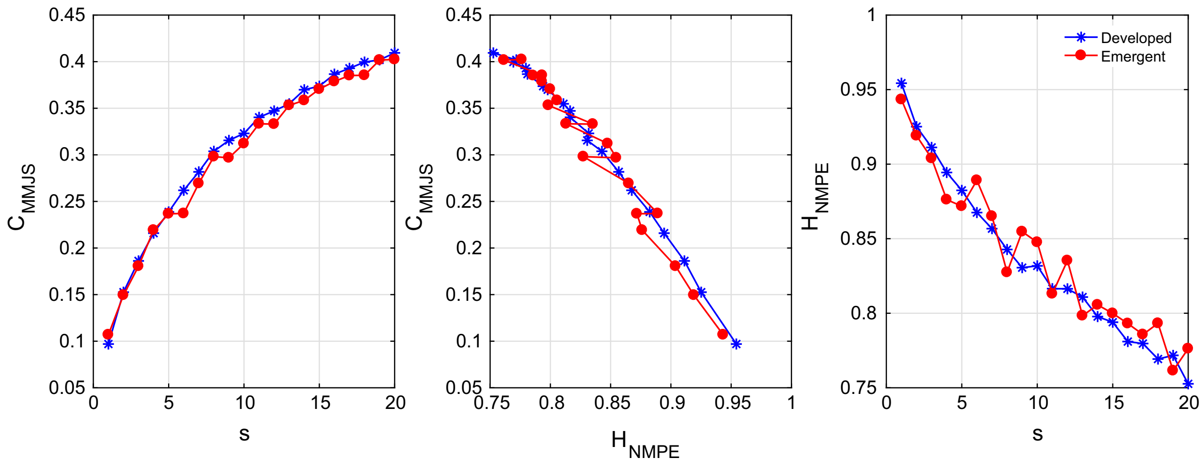

Fig. 8 Multiscale analysis of H_{NMPE} and C_{MMJS} averaged over all the stock indices of each group, developed or emergent

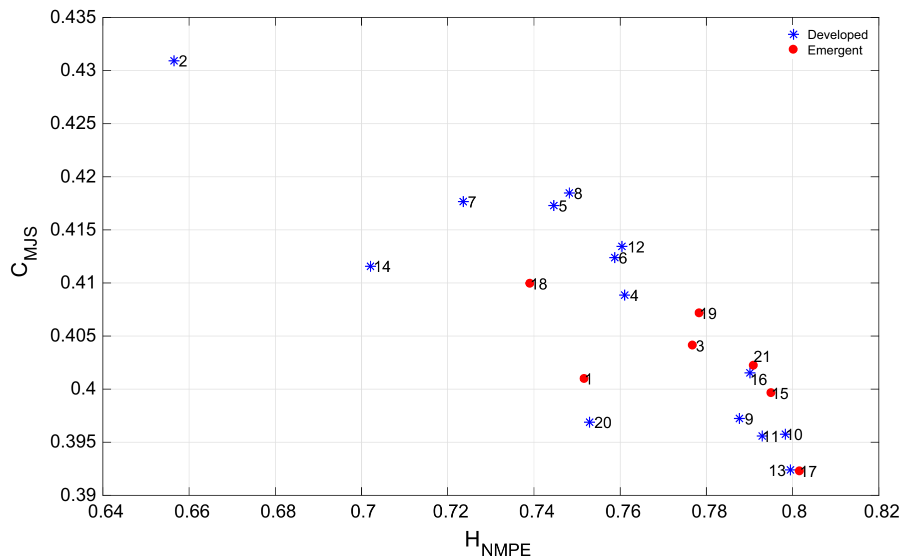

Fig. 9 Multivariate multiscale CECP at the maximum scale $s = 20$

As is given in Table 1, the 21 stock indices are divided into three different regions: seven in Americas, six in Europe and eight in Asia and Pacific. And according to MSCI methodology they are also split into two parts, stock indices from developed countries and those from emergent countries. Multiscale analysis of H_{NMPE} and C_{MMJS} averaged over all the stock indices of each region is plotted in Fig. 7. It is obvious that all the curves of C_{MMJS} grow up slowly as the scale s increases and the rate of change (slope) of three curves becomes slight gradually, implying the lower dynamical complexity at larger scales. In contrast, the curves of H_{NMPE} see an opposite trend, decreasing a little rapidly with the growth of scale; in particular, the rate of descent of Americas is higher than another two regions, which means the stronger long-range correlations. Correspondingly, the higher complexity and lower entropy are presented in multivariate multiscale CECP with the increasing scale. To deserve to be mentioned, the green curves in three graphs fluctuate a little more fiercely since there are more emergent countries in Region Asia and Pacific which is not quite stationary. By comparison, since the countries chosen in Europe are all developed, the curves of averaged values are reversely stable. To characterize the complexity and long-range correlations in stock market more comprehensively, we also plot the curves of H_{NMPE} and C_{MMJS} averaged over all the stock indices of each

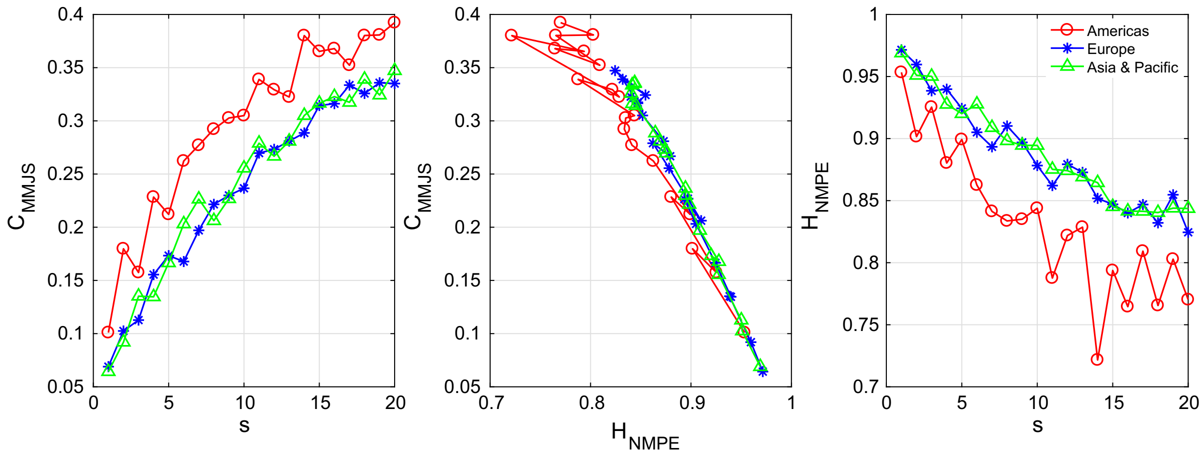

Fig. 10 Multiscale analysis of H_{NMPE} and C_{MMJS} in each region choosing three specific stock indices

group, developed or emergent in Fig. 8. Without loss of generality, the overall trends of three graphs are similar to those in Fig. 7. The most significant difference is that the related values of stock indices from emergent countries fluctuate more obviously with the increase in scale, indicating the instability in this stock market. Nevertheless, all the stock indices exhibit long-range correlations and broad fat-tailed distributions, which is shown in Fig. 9. Although the points in Fig. 9 are more scattered than those in Fig. 6, it is apparent that similar conclusions can be drawn with high-level complexity and low entropy under scrutiny.

Although the graphs above could distinguish different stock indices with different properties, the gap among them is kind of marginal. To overcome this problem, three stock indices are chosen from each region: Americas (DJI, GSPC, IXI), Europe (OMXHPI, GDAXI, OMXS30) and Asia and Pacific (SSEC, HSI, N225). Here we set the logarithmic closing price of three stock indices as the multivariate time series and then evaluate the complexity and long-range correlations among them. Figure 10 manifests the change of C_{MMJS} and H_{NMPE} with the increasing scale as well as the multivariate multiscale CECP applying improved multivariate time series. Apart from the general trends, we observe that the values of C_{MMJS} in Americas are the largest all the time and the growth rate is faster; meanwhile, the values of H_{NMPE} are the smallest and the decline rate becomes more rapid than those in other two regions. This circumstance can be explained the consistency with the efficient market hypothesis in American markets. On the other hand, the changes in other regions seem to be a little gentle by comparison.

5 Conclusions

In this paper, we extend complexity-entropy causality plane into the multivariate multiscale situation to present a deeper insight into multivariate time series and further analyze the complexity and long-range correlations of such systems. To numerically confirm the robustness and effectiveness of the proposed method, we first select seven chaotic maps, two stochastic processes and one periodic process. Note that the latter two kinds of processes are for the comparison. Most of the points related to the signals are found in the vicinity of the maximum complexity curve $C_{\max}$. And the periodic process has less complexity as expected. Furthermore, there is no doubt that the location of multivariate Gaussian white noise is close to the ideal point $(H_{\mathrm{NMPE}}, C_{\mathrm{MJS}}) = (1, 0)$. Then we study the planar representations of f^{-k} power spectrum. As the parameter k increases, the values of entropy decrease gradually, whereas the values of C_{MJS} first grow up and then decline a little, implying less randomness and higher complexity for $k > 0$. Later, to better comprehend the dynamical properties of a system, we apply multiscale approach in the analyses of multivariate CECP. In general, the irregular and unpredictable behaviors manifest in pace the growth of scale. Consequently, the complexity of almost all the chaotic maps becomes weaker in varying degrees except Rössler system. Considering that the signals in real world usually contaminated with noises, we add different levels of Gaussian white noise to four chaotic maps. Results show that the influence of noise is significant at small scales and nearly makes no sense at large scales, which also confirms the long-range constancy of systems.

Then the suggested measure is utilized in stock market for practical application. We select twenty-one stock indices and divide them into three parts and two parts according to different regions and features of country, respectively. Observe that the stock indices from emergent countries suggest more volatile than those from developed countries. And the stock indices from Region Americas have higher complexity. Nevertheless, there are long-range correlations in all the stock indices according to the multiscale analysis.

In conclusion, the proposed multivariate multiscale CECP solves the problems more efficiently involving the complexity and long-range correlations of multivariate time series.

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Compliance with ethical standards

Conflicts of interest The authors declare that they have no conflict of interest concerning the publication of this manuscript. Name: Xuegeng Mao

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